

Linearity Analysis of Traditional Single and Double Balanced Down Conversion Mixers

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Abstract

More sophisticated designs of communicating devices are required with tremendous scaling in MOS. Mixers play a vital role in all radio transmitters and receivers. To improve the sensitivity of the receivers, the low noise amplifiers are designed with very high gain. Higher gain of amplifier saturates the preceding sections and forces them to operate in the nonlinear region. MOS based Single Balanced and Double Balanced mixers are the most common designs for down conversion mixer. The paper discusses the linearity issue for both the traditional mixers. Basic definition of linearity is explained and the mathematical analysis based on Taylor's series expansion is carried out for single and double balanced mixers.

Keywords: Linearity, IIP3, mixer, mathematical analysis, amplifiers

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INTRODUCTION

With improvement in the fabrication processes, it is possible to scale the MOS device to nano meter regime. With the reduction in the device dimensions, the linearity of the device suffers, which leads to the requirement of more sophisticated circuit topologies [1]. Down conversion mixer is an important part of radio receiver, which is preceded by Low Noise Amplifier (LNA). Mixer receives two inputs, one from LNA, which is called RF signal and second from the local oscillator, which is called LO signal. The output of the mixer is known as IF or intermediate signal.

The most famous mixer topologies are single balanced and double balanced mixer topologies [2]. It is often said that the double balanced mixers are more linear compared to the single balanced one [3], but the mathematical analysis for the said comparison is not easily available anywhere. The paper discusses the definition of linearity. The mathematical analysis of linearity is presented and the same is applied to the mixers using Taylor's series expansion.

The flow of the paper is as by presenting the linearity and the mathematical formulas to find the linearity of the design, discussing single

and double balance mixer theory, presenting the linearity analysis of both the mixers. Lastly, the results and discussion and lastly the paper is concluded with remarks.

LINEARITY

MOS devices are transconductance devices. The output current change with change in the gate voltage. Due to nonlinear current-voltage relation of MOS, there may be many unwanted signals or harmonics observed in the final output, which are not applied at the input. The current voltage relation of MOS like device can be expressed using Taylor's series as:

$$y(t) = a_1x(t) + a_2x(t)^2 + a_3x(t)^3 + \dots \quad (1)$$

Where, $y(t)$ and $x(t)$ are the input to and output from the MOS device respectively and a_1 , a_2 and a_3 are the gain of respective terms. There are various effects of nonlinearity as described below.

Total Harmonic Distortion (THD)

If a sinusoid signal, $x(t) = A \cos \omega t$ is applied at the input of the MOS device, the input-output relation can be describe as shown below after neglecting the higher order term.

$$y(t) = a_1A \cos \omega t + a_2A^2 \cos^2 \omega t + a_3A^3 \cos^3 \omega t \dots \quad (2)$$

$$= a_1 A \cos \omega t + \frac{a_2 A^2}{2} (1 + \cos 2\omega t) + \frac{a_3 A^3}{4} (3 \cos \omega t + \cos 3\omega t) \dots \quad (3)$$

$$= \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4} \right) \cos \omega t + \frac{a_2 A^2}{2} \cos 2\omega t + \frac{3a_3 A^3}{4} \cos 3\omega t \dots \quad (4)$$

$$THD = \frac{\left(\frac{a_2 A^2}{2} \right)^2 + \left(\frac{3a_3 A^3}{4} \right)^2}{\left(a_1 A + \frac{3a_3 A^3}{4} \right)^2} \dots \quad (5)$$

In equation (4), the first term is the DC in the output, second term is the fundamental term with $a_1 A + \frac{3a_3 A^3}{4}$ gain, which is the amplified input signal. The remaining terms are the harmonics, which were not present in the input signal. These unwanted signals distort the required output, which is called harmonic distortion. The Total Harmonic Distortion can be defined as the ratio of the sum of the powers of all harmonic components to the power of the fundamental frequency.

Gain Compression

Normally, the signs of a_1 and a_3 in equation (4) are opposite of each other, which leads to reduction in the output. This effect compresses the expected gain at the output. At high input power levels the value of a_3 approaches a_1 . The point at which both the constant are equal the gain reduces to zero. Practically this point must never be achieved, but the reduction in the gain is always observed. The maximum reduction in the gain allowed is 1 dB as per the standards. The point where the gain of the output reduces by 1 dB compared to the expected theoretical gain is called 1 dB compression point [4].

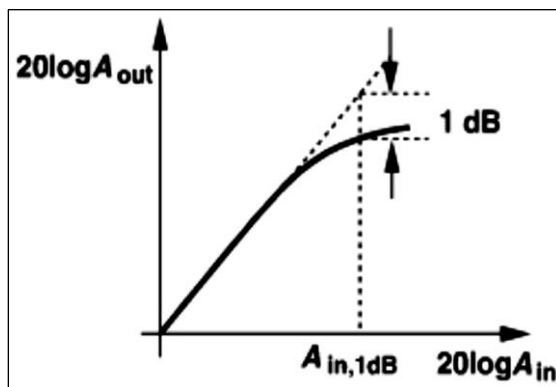


Fig. 1: 1 dB Gain Compression.

The input level must be kept below the level at which 1 dB compression point occurs to avoid the reduction in the gain. 1 dB compression is shown in Figure 1. Mathematically the amplitude of input, where the 1 dB compression occurs, is given as:

$$A_{1\text{ dB}} = \sqrt{0.145 \left| \frac{a_1}{a_3} \right|} \quad (6)$$

Inter Modulation

When more than one frequency is applied at the input of the device an interesting phenomena occurs. Assuming, $x(t) = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t$, the final output has DC term, fundamental term as [4]:

$$\omega_{\text{fundamental}}: \left(a_1 A_1 + \frac{3}{4} a_3 A_1^3 + \frac{3}{2} a_3 A_1 A_2^2 \right) \cos \omega_1 t + \left(a_1 A_2 + \frac{3}{4} a_3 A_2^3 + \frac{3}{2} a_3 A_2 A_1^2 \right) \cos \omega_2 t$$

The intermodulation product term as:

$$\frac{3a_3 A_1 A_2^2}{4} \cos(2\omega_1 + \omega_2)t + \frac{3a_3 A_1 A_2^2}{4} \cos(2\omega_1 - \omega_2)t, \frac{3a_3 A_2 A_1^2}{4} \cos(2\omega_2 + \omega_1)t + \frac{3a_3 A_2 A_1^2}{4} \cos(2\omega_2 - \omega_1)t$$

These frequency terms are called third order intermodulation (IM3) product terms. Unfortunately, the IM3 component has the same frequency as the fundamental one or they are very close to the fundamental frequency terms. If the amplitude of the IM3 terms is comparable to the amplitude of the fundamental term, then it can easily nullify the fundamental terms from the final output. The second order intermodulation products, $\omega_1 \pm \omega_2$, are also part of the output but are not in the pass band of the interested fundamental frequency terms and can be neglected. The measure of the effect of IM3 can be found using a very important parameter called input third order interception point or IIP3. It is the input point for which the amplitude of fundamental term equals the amplitude of the third order intermodulation term. The input

level equal to IIP3 is well above the 1 dB compression point, so the input is never allowed to achieve these levels. Mathematically the input amplitude for this level can be given as [4]:

$$A_{IIP3} = \sqrt{\frac{4}{3} \left| \frac{a_1}{a_3} \right|} \quad (7)$$

Figure 2 shows the representation of IIP3. Often the designers consider the THD as a measure of linearity of the circuit, but the actual measure of linearity is IIP3 of the circuit. In the next sections the famous single and double balanced mixers are discussed for their operation and then the above equations are applied to them.

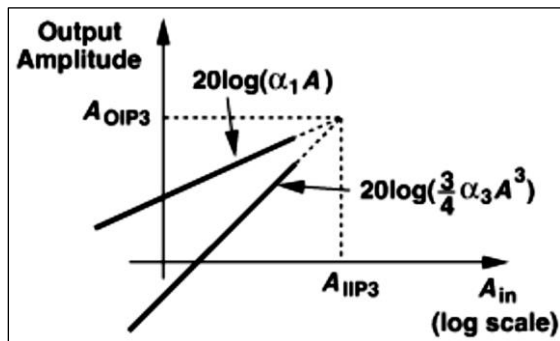


Fig. 2: Input Third Order Interception Point.

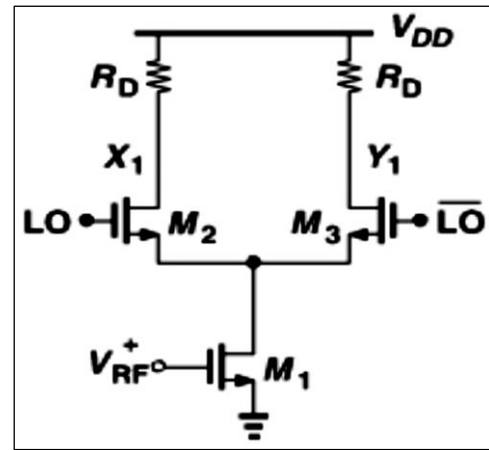
TRADITIONAL MIXERS

As discussed previously, the mixers are designed to generate the IF frequency by multiplying or mixing RF and LO signals. Due to nonlinear characteristics of MOS devices, the output of mixer contains many unwanted signals with the actual IF signal.

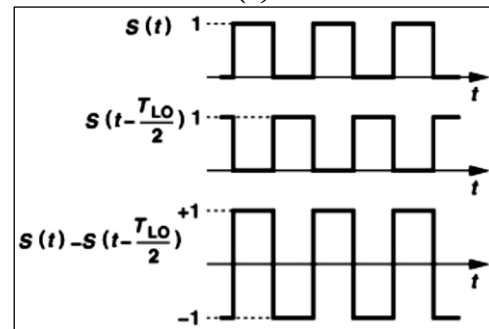
Single Balanced Mixer

Figure 3(a) shows single balanced mixer topology using MOS and Figure 3(b) shows the switching of LO signal. M1 is input transistor where input RF is applied. M1 is responsible for amplification of RF signal. Transistors M2 and M3 are switching transistors.

The switching of transistor from cutoff to saturation is performed by LO signal. M2 and M3 are given the LO signal with opposite polarity, so at a time anyone transistor is on and the other is completely off. The resistors at the drain of M2 and M3 act as load resistors.



(a)



(b)

Fig. 3: (a) Single Balance Mixer (b) LO Waveforms.

The output voltage of single balanced mixer can be expressed as:

$$V_{out} = V_{X1} - V_{Y1} \quad (8)$$

$$= V_{DD} - I_{D1}R_D - V_{DD} + I_{D2}R_D \quad (9)$$

$$= I_D R_D \left[s\left(t - \frac{T_{LO}}{2}\right) - s(t) \right] \quad (10)$$

After neglecting the higher order terms the output can be expanded as:

$$V_{out} = I_{RF} R_D \frac{4}{\pi} \left(\cos \omega_{LO} t - \frac{1}{3} \cos 3\omega_{LO} t + \frac{1}{5} \cos 5\omega_{LO} t \right) \quad (11)$$

Where, I_{RF} is $g_m V_{RF} + I_{DC}$ and g_m is the transconductance of the device and I_{DC} is the DC bias current from M1. The final output after considering only fundamental terms can be given as:

$$V_{out} = \frac{4}{\pi} I_{DC} \cos \omega_{LO} t + \frac{4}{\pi} g_m V_{RF} R_D \cos(\omega_{LO} - \omega_{RF}) t \quad (12)$$

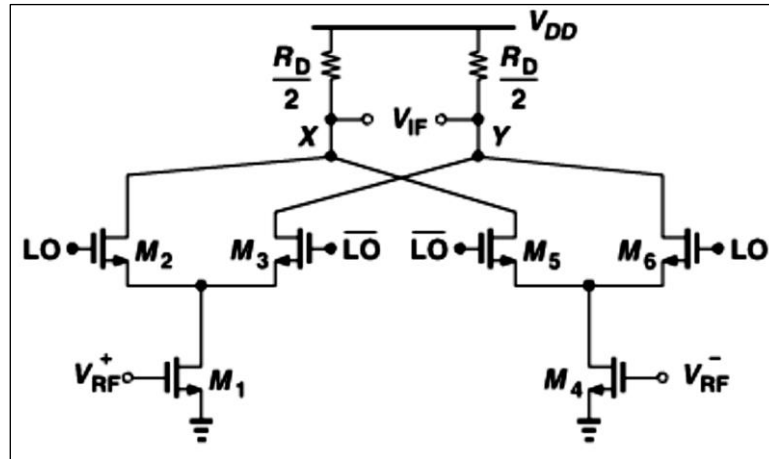


Fig. 4: Double Balance Mixer.

As it can be seen that the component of LO is also present in the output with the IF signal. This is called LO feed through. This equation is not sufficient to find out the linearity of the circuit. The linearity will be discussed in the next section.

Double Balanced Mixer

Figure 4 shows the double balanced mixer topology, which is also known as Gilbert Cell mixer.

The double balanced mixer is designed using a pair of single balanced mixer. M1, M2 and M3 make one single balanced mixer while M4, M5 and M6 makes the other. M1 and M4 are the transconductance stage which is responsible for amplification of the RF signal, while M2, M3, M5 and M6 are switching pairs. M1 and M4 are driven by the opposite polarity RF signal.

The output of double balanced mixer can be given as:

$$V_{out} = V_X - V_Y \quad (13)$$

$$\begin{aligned} V_{out} = & \frac{4}{\pi} I_{DC} \cos \omega_{LO} t \\ & + \frac{4}{\pi} g_m V_{RF} \frac{R_D}{2} \cos(\omega_{LO} - \omega_{RF}) t \\ & - \frac{4}{\pi} I_{DC} \cos \omega_{LO} t \\ & + \frac{4}{\pi} g_m V_{RF} \frac{R_D}{2} \cos(\omega_{LO} - \omega_{RF}) t \end{aligned} \quad (14)$$

$$V_{out} = \frac{4}{\pi} g_m V_{RF} R_D \cos(\omega_{LO} - \omega_{RF}) t \quad (15)$$

It is obvious from the equation that the output contains only IF component. No LO feed through is observed in the output as its single balanced counterpart. To achieve the same gain as that of single balanced mixer, double balanced mixer requires twice as much current as single balanced mixer requires. The power requirement and noise performance is also poor due to more number of transistors. These compromises are required to achieve output without RF and LO feed through.

However, as in the case of single balanced mixer, equation 14 does not give any information about the linearity performance of the mixer. In the next section the linearity analysis of both the mixers is carried out.

LINEARITY ANALYSIS

Total harmonic distortion is a part of linearity, but for LNA or mixers followed by band-pass filters, THD doesn't play major role, because the harmonics are always out of band with respect to the IF output. The major focus of linearity analysis is on IIP3 for the mixers. Even if the filters are used after mixers, the IM3 components can't be removed as they are very close to the IF frequency.

As per equations (5) (6) and (7), linearity analysis requires a_1 and a_3 values only. To find out the values, Taylor's series is considered, which is described in equation (1). It is to be noted that the system is considered as memory less. First the single balanced mixer is considered. The current-voltage relation for single balanced mixer can be given as:

$$I_D = \frac{1}{2} \mu_0 c_{ox} \frac{W}{L} \frac{(V_{GS} - V_T)^2}{1 + \left(\frac{\mu_0}{2v_{sat}L} + \theta \right) (V_{GS} - V_T)} \quad (16)$$

Where, μ_0 denotes zero field mobility for MOS, v_{sat} is saturation velocity observed by the carriers and θ is the factor indicating the effect of vertical field.

Considering $\left(\frac{\mu_0}{2v_{sat}L} + \theta \right) (V_{GS} - V_T) \ll 1$, $1 + \epsilon^{-1} = 1 - \epsilon$ can be applied to (16).

$$I_D \approx \frac{1}{2} \mu_0 c_{ox} \frac{W}{L} \left[(V_{GS} - V_T)^2 - \left(\frac{\mu_0}{2v_{sat}L} + \theta \right) (V_{GS} - V_T)^3 \right] \quad (17)$$

In (17), $V_{GS} = v_{in} + V_{GS0}$, addition of RF input and initial biasing to keep the transistor in saturation. Moreover, $\frac{1}{2} \mu_0 c_{ox} \frac{W}{L}$ is replaced by K and $\left(\frac{\mu_0}{2v_{sat}L} + \theta \right)$ is replaced by a to simplify the equation and the final relation can be represented as:

$$I_D \approx K[2 - 3a(V_{GS0} - V_T)](V_{GS0} - V_T)v_{in} + K[1 - 3a(V_{GS0} - V_T)]v_{in}^2 - K a v_{in}^3 + K(V_{GS0} - V_T)^2 - aK(V_{GS0} - V_T)^3 \quad (18)$$

Comparing (18) with (1) gives,

$$a_1 = [2 - 3a(V_{GS0} - V_T)](V_{GS0} - V_T),$$

$$a_3 = -Ka$$

Which lead to the value of IIP3 as:

$$A_{IIP3} = \sqrt{\frac{\frac{8}{3}(V_{GS0} - V_T)}{\frac{\mu_0}{2v_{sat}L} + \theta} - 4(V_{GS0} - V_T)^2} \quad (19)$$

Which can achieve the maximum value of $\frac{2}{3} \frac{1}{\frac{\mu_0}{2v_{sat}L} + \theta}$.

For double balanced mixer the current equation can be written as:

$$I_{D1} - I_{D2} \approx \frac{1}{2} \mu_0 c_{ox} \frac{W}{L} v_{in} \sqrt{\frac{4I_{SS}}{\mu_0 c_{ox} \frac{W}{L}} - v_{in}^2} \quad (20)$$

For $v_{in} \ll \frac{I_{SS}}{\mu_0 c_{ox} \frac{W}{L}}$, (20) can be expanded as:

$$I_{D1} - I_{D2} \approx \frac{1}{2} \mu_0 c_{ox} \frac{W}{L} v_{in} \sqrt{\frac{4I_{SS}}{\mu_0 c_{ox} \frac{W}{L}}} \left(1 - \frac{1}{2} \frac{v_{in}^2}{\frac{4I_{SS}}{\mu_0 c_{ox} \frac{W}{L}}} \right) \quad (21)$$

Which leads to IIP3 as [5]:

$$A_{IIP3} = 4 \sqrt{\frac{2}{3}} (V_{GS0} - V_T) \quad (22)$$

Which is maximum for maximum value of overdrive. The same equations can be applied to find the THD for both mixers and can be written as:

$$THD_{Single Balanced} = \frac{V_m}{4(V_{GS0} - V_T)} \quad (23)$$

$$THD_{Double Balanced} = \frac{V_m^2}{32(V_{GS0} - V_T)^2} \quad (24)$$

It must be noted that in the analysis presented so far only amplifying transistors or the transconductance stage is considered. The switching pairs are either completely off or completely on, so their contribution in the analysis of linearity can be ignored compared to the transconductance stage. So, single balanced mixer topology can be considered as simple single stage amplifier. The double balanced mixer topology can be considered as differential amplifier. The equation discussed in the paper is equally applicable to amplifiers as well.

RESULTS

Table 1 displays the Compression point and IIP3 comparisons of single and double balanced mixer. The results are obtained using spice.

Table 1: Simulated Linearity Comparison.

	Single Balanced	Double Balanced
RF	2.4 GHz	2.4 GHz
IF	250 MHz	250 MHz
Supply	1.8 V	1.8 V
Technology	0.18 μ	0.18 μ
P – 1dB	-15.7 dBm	-12.46 dBm
IIP3	-2.53 dBm	1.85 dBm

It can be seen from equations (23) and (24) that as far as THD is concern double balanced mixer has better performance. The IIP3, which is the main linearity issue, is also better in case of double balanced mixer. In single balanced mixer only one transistor is used, so all the harmonics are present in the output. While in the double balanced mixer due to differential topology even order harmonics get cancelled out which improve the THD performance. The IIP3 of double balanced mixer is higher compared to the single balance counterpart. Due to better IIP3 double balance mixer also offers better 1 dB compression point.

CONCLUSION

Mathematic expressions of linearity is presented and applied to the single and double balanced mixers. Detailed analysis of linearity for both the mixer topology is carried out. The overall linearity of double balanced mixer is found better compared to the single balanced mixer. The double balanced mixer offers better THD performance as well due to cancellation of even harmonics. The simulation for the same is carried out using 0.18 μ technology and 1.8 V supply in spice.

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